

$$i) \int k \, dx = kx + C$$

~~proof~~
proof

$$ii) \int x^n \, dx = \frac{1}{n+1} x^{n+1} + C.$$

$$iii) \int k f(x) \, dx \text{ iff } k \int f(x) \, dx$$

$$iv) \int (f(x) \pm g(x)) \, dx = \int f(x) \, dx \pm \int g(x) \, dx$$

$$pf) - \frac{d}{dx} (kx + C) = k. \quad \therefore \int k \, dx = kx + C$$

$$- \frac{d}{dx} \left(\frac{1}{n+1} x^{n+1} + C \right) = \frac{1}{n+1} \cdot (n+1) x^n = x^n$$

$$- \frac{d}{dx} \left(k \int f(x) \, dx \right) = k \cdot \frac{d}{dx} \left(\int f(x) \, dx \right) = k f(x)$$

$$\therefore k \int f(x) \, dx = \int k f(x) \, dx$$

$$- \frac{d}{dx} \left(\int f(x) \pm g(x) \right) = \frac{d}{dx} \left(\int f(x) \, dx \right) \pm \frac{d}{dx} \left(\int g(x) \, dx \right) = f(x) \pm g(x)$$

$$\therefore \int (f(x) \pm g(x)) \, dx = \int f(x) \, dx \pm \int g(x) \, dx$$